

NELSON LAGOON, AK, USA

WMO: 703058

Lat: 56.010N

Lon: 161.160W

Elev: 4

StdP: 101.27

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 00110

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-13.0	-12.2	-17.2	0.8	-13.5	-15.0	1.0	-11.8	17.6	-1.8	15.9	-2.6	6.4	330	0.874

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.2	16.1	14.1	15.0	13.4	14.0	12.6	14.5	15.6	13.7	14.7	12.9	13.8	6.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	9.9	14.9	13.0	9.3	14.0	12.4	9.0	13.5	40.4	15.6	38.3	14.8	36.4	13.9	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 16.1	14.5	12.7	DB	-15.3	19.2	4.5	1.8	-18.6	20.5	-21.2	21.5	-23.8	22.5	-27.1	23.8	(4)
(5)			WB	-15.0	17.2	5.2	2.1	-18.8	18.7	-21.9	20.0	-24.8	21.2	-28.6	22.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-2.2	-1.2	-2.3	0.9	4.5	7.9	10.6	11.7	9.9	6.1	2.2	0.2
	DBStd	5.86	4.62	4.08	4.80	3.22	2.74	2.19	2.10	1.86	2.01	2.51	2.94	3.37
	HDD10.0	2294	379	313	382	274	171	69	18	3	24	124	233	304
	HDD18.3	5214	638	546	640	524	429	313	241	205	252	381	483	562
	CDD10.0	122	0	0	0	0	0	6	36	56	22	2	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.9	6.8	5.9	6.4	6.4	4.8	4.8	4.4	4.9	6.5	6.2	6.5	7.3
Precipitation	PrecAvg	1051	98	90	72	58	66	75	93	76	113	97	102	111
	PrecMax	1430	160	185	148	95	125	125	146	113	183	166	167	239
	PrecMin	789	40	29	21	19	17	32	37	28	75	49	38	41
	PrecStd	155	32	37	33	21	25	23	27	21	27	31	33	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	6.3	7.0	8.7	11.3	14.9	17.4	18.4	16.6	12.6	8.5	7.7
		MCWB	4.4	4.8	5.5	6.6	8.8	12.7	14.6	15.9	14.6	10.7	7.3	6.7
	2%	DB	4.5	5.1	5.1	7.1	9.8	13.0	15.7	16.3	14.7	11.6	7.7	6.2
		MCWB	3.5	3.9	3.9	5.3	7.8	11.2	13.5	14.4	13.0	9.9	6.7	5.1
	5%	DB	3.7	4.2	4.0	6.0	8.9	11.8	14.4	15.2	13.6	10.9	7.0	5.1
		MCWB	2.8	3.2	2.9	4.5	7.3	10.3	12.7	13.9	12.0	9.4	6.0	4.0
	10%	DB	2.9	3.4	3.0	4.9	7.9	10.8	13.4	14.3	12.7	9.7	6.1	4.1
		MCWB	2.1	2.4	2.1	3.7	6.5	9.5	12.1	13.1	11.2	8.3	5.1	3.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	5.3	5.6	7.0	10.1	13.0	15.1	16.4	14.9	11.2	7.9	6.8
		MCDB	5.2	6.1	6.7	8.6	10.9	14.3	16.7	17.1	16.2	12.0	8.1	7.4
	2%	WB	3.8	4.2	4.1	5.5	8.5	11.6	13.9	14.8	13.4	10.3	7.0	5.3
		MCDB	4.3	4.8	4.8	6.7	9.2	12.6	15.2	15.9	14.5	11.3	7.4	6.0
	5%	WB	3.0	3.3	3.0	4.6	7.7	10.6	13.0	14.1	12.3	9.5	6.1	4.2
		MCDB	3.5	4.0	3.6	5.8	8.6	11.6	14.1	15.0	13.3	10.7	6.8	5.0
	10%	WB	2.3	2.5	2.3	3.8	6.9	9.6	12.4	13.3	11.4	8.3	5.2	3.4
		MCDB	2.9	3.3	3.0	4.8	7.8	10.6	13.3	14.2	12.4	9.4	6.1	4.0
Mean Daily Temperature Range	5% DB	MDBR	3.6	3.9	4.2	3.7	3.6	3.4	3.4	3.2	3.0	3.3	3.5	3.4
		MCDBR	3.5	4.2	4.0	4.5	4.7	5.1	5.1	4.5	3.9	3.7	3.6	3.5
		MCWBR	3.0	3.6	3.3	3.4	3.3	3.5	3.5	3.4	3.2	2.9	3.4	3.2
	5% WB	MCDBR	3.4	4.1	4.0	4.2	4.4	4.8	4.8	4.4	3.8	3.5	3.5	3.4
		MCWBR	3.2	3.7	3.5	3.4	3.4	3.6	3.6	3.4	3.3	3.1	3.5	3.2
Clear-Sky Solar Irradiance	taub	0.288	0.327	0.340	0.387	0.390	0.385	0.377	0.389	0.364	0.340	0.322	0.280	
	taud	2.308	2.293	2.270	2.183	2.269	2.372	2.451	2.389	2.399	2.325	2.173	2.321	
	Ebn at Noon	626	732	819	826	849	859	860	821	789	703	540	546	
	Edh at Noon	51	77	99	126	123	113	102	102	88	73	58	40	
All-Sky Solar Radiation	RadAvg	0.47	1.00	2.34	3.68	4.48	4.75	4.11	3.37	2.44	1.38	0.59	0.32	
	RadStd	0.06	0.12	0.38	0.28	0.42	0.49	0.43	0.31	0.22	0.11	0.06	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	+1.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page